

The Rung Family of Lonely Runner Spectra

$$\text{gap}(1, 2, \dots, n-1, kn) = k/(kn+1)$$

An elementary proof with complete argmax structure, the closed-form filler characterization $X(S)$, and the exhaustive $n = 8$ census at $V = 50$ — 536,878,650 speed sets, exact arithmetic throughout.

MIKEAA2020

lonely-runner project · computational-check
companion to the audit “rung-2 ladder and R1”

Unique argmax $\pm k/(kn+1)$ for every $k \geq 2$ — no coprimality condition (no-wrap induction)
Filler theorem — kill & touch, with an explicit radius $R(S)$; verified on 1,092 cores
Inter-rung window $(1/9, 2/17)$ **empty** at $V \leq 50$; tight and rung-2 zoos closed

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1 Introduction

For a finite set V of positive integer speeds, the *loneliness* of V is the quantity

$$\text{gap}(V) = \sup_{t \in \mathbb{R}} \min_{v \in V} \|vt\|,$$

where $\|x\|$ denotes the distance of x to the nearest integer. The Lonely Runner Conjecture (LRC) asserts that every m -element set of distinct speeds satisfies $\text{gap}(V) \geq \frac{1}{m+1}$, and the bound is tight for the consecutive family $V = \{1, 2, \dots, m\}$, for which $\text{gap}(V) = \frac{1}{m+1}$ exactly. The conjecture is known for up to seven speeds and open from eight onward; exhaustive computational verification is therefore one of the few systematic tools available, and the structure of the *spectrum* — the multiset of gap values realized by all m -sets up to some radius — is a recurring theme of the computational literature.

This note is a self-contained study of one spectral family. For $n \geq 2$ and $k \geq 1$ define the *rung family*

$$F(n, k) = \{1, 2, \dots, n-1\} \cup \{kn\},$$

an n -element set consisting of the first $n-1$ consecutive speeds together with the single large speed kn . The rung $k=1$ is the classical consecutive family. The values

$$\frac{1}{n+1}, \frac{2}{2n+1}, \frac{3}{3n+1}, \dots, \frac{k}{kn+1}$$

appear as a rigid arithmetic progression at the very bottom of every computed spectrum — the “rungs” of a ladder — and the purpose of this note is to prove that they are exact, to determine the complete maximizer structure behind them, and to show how the same structure governs the neighbouring zoo of extremal sets.

Three theorems are proved. **Theorem 2.2** (Section 3) gives, for every $n \geq 2$ and $k \geq 1$, the exact value $\text{gap}(F(n, k)) = \frac{k}{kn+1}$ together with the *complete* argmax structure: for $k \geq 2$ the maximum is attained at the single pair $t = \pm \frac{k}{kn+1} \pmod{1}$, the binding runners are exactly $\{1, kn\}$, and every other runner sits at distance at least $\frac{k+1}{kn+1}$; for $k=1$ the maximizers are exactly the unit fractions $a/(n+1)$, a structural contrast that explains why the consecutive family is exceptional. The upper bound is an elementary pigeonhole argument that generalizes the proof of the $k=2$ case in the project’s earlier audit; the equality case is new in one decisive respect: the uniqueness of the maximizer is proved by a *no-wrap induction* that requires no coprimality hypothesis, closing the gcd-degenerate case that the earlier sum-based argument left open. **Theorem 2.4** (Section 6) is a closed-form characterization of the *fillers* of a core: for a speed set S whose gap exceeds a rational target g , the set $X(S) = \{x : \text{gap}(S \cup \{x\}) = g\}$ is characterized exactly by two finitely checkable conditions — a window condition on each escape interval of the core and a cleanliness condition at one level time — and is finite with an explicit a priori radius. This turns the empirically observed ladder structure of extremal sets into a theorem. Finally, Section 7 reports an exhaustive census of all $\binom{50}{8} = 536,878,650$ eight-element speed sets with speeds up to 50, carried out with a filtered exact solver whose certificate is a one-line consequence of the definition of gap: the census shows the tight zoo closed, the rung-2 zoo closed in primitive structure, and — the target question of the hunt — the inter-rung window $(\frac{1}{9}, \frac{2}{17})$ empty.

All computations use exact integer or rational arithmetic throughout; no floating point is involved anywhere. The machine checks summarized in Section 8 were run with two independent implementations of the gap computation that must agree on every value.

2 Definitions and statements

Throughout, $\|x\| = \min_{m \in \mathbb{Z}} |x - m|$ is the distance to the nearest integer, so that $\|x\| \in [0, \frac{1}{2}]$ and $\|-x\| = \|x\|$. Two elementary facts are used repeatedly: $\|qx\| \leq q\|x\|$ for every integer

$q \geq 1$; and for a rational $t = a/d$ in lowest terms, $\|v \cdot a/d\|$ is computed exactly as $\min(r, d-r)/d$ with $r = av \bmod d$.

Definition 2.1 (gap, argmax, binders). For a finite set V of positive integers,

$$\text{gap}(V) = \sup_{t \in \mathbb{R}} \min_{v \in V} \|vt\|,$$

where the supremum may be taken over $t \in [0, 1]$. The *argmax set* $\text{Argmax}(V)$ is the set of $t \in [0, 1)$ attaining the supremum. For t in the argmax set, the *binders at t* are the speeds $v \in V$ with $\|vt\| = \text{gap}(V)$.

The function $g_V(t) = \min_{v \in V} \|vt\|$ is continuous and piecewise linear, with breakpoints at the peaks $t = \text{odd}/(2v)$ and crossings $t(v_i \pm v_j) \in \mathbb{Z}$; its maximum is attained among these, which is what the exact solvers enumerate. The Lonely Runner Conjecture in this normalization states $\text{gap}(V) \geq \frac{1}{|V|+1}$ for $|V|$ distinct speeds, with equality for the consecutive family.

The three main results are as follows.

Theorem 2.2 (rung family, value and complete argmax structure). *Let $n \geq 2$ and $k \geq 2$, and put $D = kn + 1$ and $\delta = k/D$. Then:*

- (i) $\text{gap}(\{1, \dots, n-1, kn\}) = \frac{k}{kn+1}$;
- (ii) $\text{Argmax} = \left\{ \frac{k}{kn+1}, \frac{kn+1-k}{kn+1} \right\}$, *i.e. the maximum is attained at exactly two times, reflections of each other;*
- (iii) *at $t = \delta$ the binders are exactly the runners 1 and kn , both at distance δ ;*
- (iv) *every other runner j , $2 \leq j \leq n-1$, has $\|j\delta\| \geq \frac{k+1}{kn+1} > \delta$.*

No coprimality condition on (n, k) is required.

Theorem 2.3 (base rung $k = 1$). *For $n \geq 2$, $\text{gap}(\{1, 2, \dots, n\}) = \frac{1}{n+1}$, and the argmax set is exactly*

$$\left\{ a/(n+1) : 0 \leq a \leq n, \gcd(a, n+1) = 1 \right\},$$

the grid of units modulo $n+1$; at each such time $t = a/(n+1)$ the binders are the speeds j with $ja \equiv \pm 1 \pmod{n+1}$.

Thus the rung family exhibits a structural dichotomy: the base rung has $\varphi(n+1)$ maximizers spread over the units grid, while every higher rung collapses to a single maximizer pair. The proof of Theorem 2.2 will show that the collapse is caused by exactly one inequality, $D \geq 2n+1$, which prevents the residue walk of the equality configuration from wrapping around the modulus.

Theorem 2.4 (closed-form filler characterization). *Let S be a finite set of positive integers, $g \in (0, \frac{1}{2})$ rational, and $g_S(t) = \min_{s \in S} \|st\|$. Define*

$$E(S) = \{t \in [0, 1] : g_S(t) > g\}, \quad L(S) = \{t \in [0, 1] : g_S(t) = g\},$$

the escape set and the level set of the core. Then $E(S)$ is a finite union of open intervals (α_j, β_j) with $0 < \alpha_j < \beta_j < 1$ and $g_S(\alpha_j) = g_S(\beta_j) = g$, and $L(S)$ is a finite set containing $\{\alpha_j, \beta_j\}_j$. For an integer $x \geq 1$ with $x \notin S$,

$$\text{gap}(S \cup \{x\}) = g \iff (A) \wedge (B),$$

where

- (A) for every escape interval (α_j, β_j) there is an integer m_j with $m_j - g \leq x\alpha_j$ and $x\beta_j \leq m_j + g$ (equivalently, $x(\beta_j - \alpha_j) \leq 2g$ and the interval $[x\beta_j - g, x\alpha_j + g]$ contains an integer);
- (B) some $t \in L(S)$ satisfies $\|xt\| \geq g$.

Moreover, if $\text{gap}(S) > g$ then (A) forces $x \leq 2g/\max_j(\beta_j - \alpha_j)$, so the filler set

$$X(S) := \{x \geq 1 : \text{gap}(S \cup \{x\}) = g\}$$

is finite with the explicit radius $R(S) = \lfloor 2g/\max_j(\beta_j - \alpha_j) \rfloor$; if $\text{gap}(S) < g$ then $X(S) = \emptyset$; and if $\text{gap}(S) = g$ then $X(S) = \{x : (B)\}$ is infinite.

Condition (A) says that the filler must *kill* every escape of the core: wherever the core exceeds the target, the filler's sawtooth must be at most g , which forces the image $x \cdot (\alpha_j, \beta_j)$ to lie inside a single window $[m_j - g, m_j + g]$ of length $2g$ on the circle $\mathbb{R}/\mathbb{Z}$. Condition (B) says the filler must be *clean* (at distance $\geq g$) at some level time of the core, which is precisely where the extension attains the value g : a filler can bind only at a level time of the core. Together the two conditions reduce the search for all fillers to a finite, exactly checkable list of two-sided congruence inequalities, computed once from the core's own geometry.

The proofs occupy Sections 3–6, the census of Section 7 supplies the computational picture at $n = 8$, and Section 8 summarizes the machine verification of every statement in exact arithmetic.

3 The upper bound

Lemma 3.1 (pigeonhole step). *Let $n \geq 2$, $k \geq 2$, $D = kn + 1$, $\delta = k/D$. For every real t ,*

$$\min(\|t\|, \|2t\|, \dots, \|(n-1)t\|, \|knt\|) \leq \delta.$$

Proof. Suppose, for contradiction, that $\|jt\| > \delta$ for $1 \leq j \leq n-1$ and $\|knt\| > \delta$, and consider the $n+1$ points $P = \{0, t, 2t, \dots, nt\}$ on the circle $\mathbb{R}/\mathbb{Z}$.

Classical step. Among $n+1$ points on the unit circle, two lie at circular distance at most $\frac{1}{n+1}$. Their index difference d lies in $\{1, \dots, n\}$, so $\|dt\| \leq \frac{1}{n+1}$. Since

$$\frac{1}{n+1} < \frac{k}{kn+1} = \delta \iff k(n+1) > kn+1 \iff k > 1,$$

and $\|jt\| > \delta$ for all $j \leq n-1$, the difference must be $d = n$: writing $\beta = \|nt\|$,

$$\beta \leq \frac{1}{n+1} < \delta.$$

Adjacency. For $0 \leq i < j \leq n$ with $(i, j) \neq (0, n)$, the index difference $j - i$ lies in $\{1, \dots, n\}$ and, whenever $j - i \leq n-1$, is the index of a runner, so the circular distance of it and jt equals $\|(j-i)t\| > \delta$. The only pair with $j - i = n$ is $(0, n)$ itself. Hence $\{0, nt\}$ is the unique pair at distance below δ , and 0 and nt are adjacent in the circular order: a point z strictly inside the short arc from 0 to nt would satisfy $\|z\| < \beta < \delta$, but $z = jt$ with $1 \leq j \leq n-1$, a contradiction.

Gap counting. The $n+1$ circular gaps of P consist of the gap β (between 0 and nt) and n further gaps, each bounded below by the circular distance of its endpoint pair, which is a pair other than $\{0, nt\}$ and hence has distance $> \delta$. Summing to 1,

$$1 > \beta + n\delta = \beta + \frac{nk}{kn+1} \implies \beta < \frac{1}{kn+1} = \frac{\delta}{k}.$$

Kill. Finally $\|knt\| = \|k \cdot nt\| \leq k\|nt\| = k\beta < k \cdot \frac{\delta}{k} = \delta$, contradicting $\|knt\| > \delta$. $\square$

The lower bound half of Theorem 2.2(i) is a direct evaluation at $t = \delta$: $\|1 \cdot \delta\| = \delta$, and

$$kn \cdot k = k(kn) \equiv k \cdot (-1) = -k \pmod{D},$$

since $kn \equiv -1 \pmod{D}$, whence $\|kn\delta\| = k/D = \delta$ (because $2k < kn + 1$ for $n \geq 2$). So $\text{gap}(F(n, k)) \geq \delta$, and Lemma 3.1 gives equality. This proves Theorem 2.2(i) for all $k \geq 2$. For $k = 1$ the classical step already suffices: $d \in \{1, \dots, n\}$ are *all* runners when $k = 1$ (the speed n is present), so $\|dt\| \leq \frac{1}{n+1}$ directly; and $t = \frac{1}{n+1}$ attains $\frac{1}{n+1}$ through runners 1 and n . This is the value half of Theorem 2.3.

4 The equality case: no-wrap induction

Throughout this section $n \geq 2$, $k \geq 2$, $D = kn + 1$, $\delta = k/D$, and t is a time at which the minimum equals δ , i.e. $\|jt\| \geq \delta$ for $1 \leq j \leq n-1$ and $\|knt\| \geq \delta$. The analysis has four steps, and it is the fourth that removes the coprimality hypothesis used in earlier treatments.

Lemma 4.1 (gap multiset). *In the equality case, the circular gaps of $P = \{0, t, \dots, nt\}$ are exactly $\{\frac{1}{D}, \frac{k}{D}, \dots, \frac{k}{D}\}$ (n copies), the gap of length $\frac{1}{D}$ lying between 0 and nt .*

Proof. Repeating the classical step with the weak hypothesis gives $\beta := \|nt\| \leq \frac{1}{n+1} < \delta$, so $\{0, nt\}$ is still the unique pair below δ and $0, nt$ are adjacent. The other n gaps are each at least the circular distance of their endpoint pair, which is a runner difference of index in $\{1, \dots, n-1\}$, hence $\geq \delta$. If $\beta < \frac{1}{D}$, then $\|knt\| \leq k\beta < \frac{k}{D} = \delta$ contradicts the hypothesis; so $\beta = \frac{1}{D}$. The n remaining gaps then sum to $1 - \frac{1}{D} = \frac{nk}{D} = n\delta$, each being $\geq \delta$, so all equal $\delta = \frac{k}{D}$. $\square$

Scaling by D , the points of P in circular order are partial sums of a sequence of $n+1$ steps consisting of n steps of size k and one step of size 1 (the β -step between nt and 0). If the unit step occupies position $p+1$ in the circular order, $0 \leq p \leq n$, then

$$P = \left\{ \frac{jk}{D} : 0 \leq j \leq p \right\} \cup \left\{ \frac{jk+1}{D} : p \leq j \leq n-1 \right\}, \quad (1)$$

the case $p = n$ being the pure arithmetic progression and $p = 0$ the reflected one.

Lemma 4.2 (no mixed configurations). $p \in \{0, n\}$.

Proof. Suppose $1 \leq p \leq n-1$. Then both $\frac{pk}{D}$ and $\frac{pk+1}{D}$ belong to P by (1), and they are distinct points at circular distance $\frac{1}{D} < \delta$. Being two points of P , they are it and jt for some indices $0 \leq i, j \leq n$, whose index difference lies in $\{1, \dots, n\}$. If $|i-j| \leq n-1$, then $\|(i-j)t\| = \frac{1}{D} < \delta$ contradicts $\|(i-j)t\| \geq \delta$ (the difference is a runner). Hence $|i-j| = n$, so $\{i, j\} = \{0, n\}$ and the two points are 0 and nt . But $0 < pk < D$ and $0 < pk+1 \leq (n-1)k+1 < D$, so neither pk nor $pk+1$ is 0 modulo D : neither point is 0, a contradiction. $\square$

Lemma 4.3 (no-wrap induction). *Let $D = kn + 1$ with $k \geq 2$, and let $m \in \{1, \dots, n\}$ satisfy*

$$\{jm \bmod D : 0 \leq j \leq n\} = \{0, 1, \dots, n\}. \quad (2)$$

Then $m = 1$.

Proof. Write $\sigma(j) = jm \bmod D \in \{0, \dots, n\}$ for $0 \leq j \leq n$; the hypothesis (2) says precisely that these are the residues of the walk and that all of them land in $\{0, \dots, n\}$. Now $\sigma(0) = 0$, and for each j ,

$$\sigma(j+1) \equiv \sigma(j) + m \pmod{D}.$$

Both $\sigma(j+1)$ and $\sigma(j) + m$ lie in $\{0, \dots, D-1\}$: the former by definition, the latter because

$$\sigma(j) + m \leq n + n = 2n \leq D-1,$$

using $D = kn + 1 \geq 2n + 1$. Two numbers in $\{0, \dots, D - 1\}$ that are congruent modulo D are equal, so $\sigma(j + 1) = \sigma(j) + m$ *without reduction*, for every j . By induction $\sigma(j) = jm$ as integers, so $jm \leq n$ for all $j \leq n$; taking $j = n$ gives $nm \leq n$, i.e. $m \leq 1$, and $m \geq 1$ finishes the proof. $\square$

Proof of Theorem 2.2. Part (i) was proved at the end of Section 3. For (ii)–(iv), let t be a time attaining the minimum δ . By Lemmas 4.1 and 4.2, P is either the pure progression $\{\frac{jk}{D} : 0 \leq j \leq n\}$ ($p = n$) or its reflection ($p = 0$). In the pure case $t \equiv \frac{mk}{D} \pmod{1}$ for some $m \in \{1, \dots, n\}$ (as t is itself a point of P), and the point set equality $\{jt \bmod 1\} = P$ reads

$$\{jmk \bmod D : 0 \leq j \leq n\} = \{jk \bmod D : 0 \leq j \leq n\};$$

multiplying by the unit $k^{-1} \bmod D$ (note $\gcd(k, kn + 1) = 1$) yields (2) of Lemma 4.3, so $m = 1$ and $t = \frac{k}{D} = \delta$. In the reflected case, $P = \{\frac{jk+1}{D} : 0 \leq j \leq n - 1\}$; applying the pure case to $1 - t$ (which has the same gap value, since $\|j(1 - t)\| = \|jt\|$, and whose point set is $\{j(1 - t)\} = -\{jt\} = \{\frac{jk}{D} : 1 \leq j \leq n\} \cup \{0\}$, the pure progression) yields $1 - t = \delta$, i.e. $t = 1 - \delta = \frac{D-k}{D}$. This proves (ii).

For (iii) and (iv), evaluate at $t = \delta$: $\|\delta\| = \delta$ (runner 1); $\|kn\delta\| = \|-k/D\| = \delta$ (runner kn); and for $2 \leq j \leq n - 1$,

$$\|j\delta\| = \frac{\min(jk, D - jk)}{D} \geq \frac{\min(2k, k + 1)}{D} = \frac{k + 1}{kn + 1} > \delta,$$

since $jk \geq 2k \geq k + 1$ and $D - jk \geq D - (n - 1)k = k + 1$. $\square$

Remark 4.4 (why the gcd condition disappears). The earlier treatment of the $k = 2$ case — and its immediate generalization — derived (2) and then summed the least residues, obtaining $(m - 1)n(n + 1)/2 = (kn + 1)Q$; concluding $m = 1$ required $\gcd(n(n + 1)/2, kn + 1) = 1$, which always holds for $k = 2$ but can fail for $k \geq 3$ (e.g. $n = k = 3$, where the gcd is 2). Lemma 4.3 replaces the sum argument by the observation that the residue walk σ never wraps, which is where $k \geq 2$ — and only that — is used. As a complement, the arithmetic obstruction itself is harmless: since $\gcd(n, kn + 1) = 1$, one always has $\gcd(n(n + 1)/2, kn + 1)$ dividing $n + 1$, hence dividing $k(n + 1) - (kn + 1) = k - 1$; so the degenerate modulus can only be a divisor of $k - 1$, and for $k \in \{2, 3\}$ the sum argument already suffices for all n . The no-wrap induction covers all $k \geq 2$ uniformly.

Remark 4.5 (the base rung is different). For $k = 1$ one has $D = n + 1 \leq 2n$, so the inequality $\sigma(j) + m \leq D - 1$ fails — and indeed the conclusion fails: by Theorem 2.3 there are $\varphi(n + 1)$ maximizers, not two. The rung family thus exhibits a sharp dichotomy at $k = 1$ versus $k \geq 2$, governed precisely by whether $D \geq 2n + 1$.

5 The base rung

Proof of Theorem 2.3. The value was proved in Section 3. Let t attain the minimum $\frac{1}{n+1} =: \delta$ for the speed set $\{1, \dots, n\}$, so $\|jt\| \geq \delta$ for all $1 \leq j \leq n$. The classical step gives $\beta := \|nt\| \leq \frac{1}{n+1} = \delta$, and $0, nt$ are adjacent. But now runner n is present in the speed set, so $\beta = \|nt\| \geq \delta$ at the argmax; hence $\beta = \delta$ exactly, and the n other gaps, each $\geq \delta$ and summing with β to $1 = (n + 1)\delta$, all equal δ as well. So all $n + 1$ circular gaps of P equal $\delta = \frac{1}{n+1}$: P is the *equally spaced* set $\{0, \frac{1}{n+1}, \frac{2}{n+1}, \dots, \frac{n}{n+1}\}$.

Hence $t = \frac{m}{n+1}$ for some $m \in \{1, \dots, n\}$, and the point set equality $\{jt \bmod 1\} = P$ becomes

$$\{jm \bmod (n + 1) : 0 \leq j \leq n\} = \{0, 1, \dots, n\},$$

which holds if and only if m is a unit modulo $n + 1$: if $c = \gcd(m, n + 1) > 1$, every residue jm is divisible by c and the right-hand side cannot be covered. Conversely, for a unit m the

map $j \mapsto jm$ permutes the residues. At $t = \frac{a}{n+1}$ with a a unit, the value is $\min_j \|\frac{ja}{n+1}\| = \frac{1}{n+1}$, attained exactly by the speeds j with $ja \equiv \pm 1 \pmod{n+1}$ (which exist since a is invertible). $\square$

The two theorems together give the rung ladder its exact shape. In every computed spectrum the bottom values are precisely the rung values $\frac{k}{kn+1}$ present at the given radius: for $n = 7$ the ladder reads $\frac{1}{8}, \frac{2}{15}, \frac{3}{22}, \frac{4}{29}, \frac{5}{36}, \frac{6}{43}, \frac{7}{50}$ and for $n = 8$ it reads $\frac{1}{9}, \frac{2}{17}, \frac{3}{25}, \frac{4}{33}, \frac{5}{41}, \frac{6}{49}$ — each realized by the family member $F(n, k)$ itself, and, as Section 7 shows, each carrying its own zoo of further extremal sets.

6 The filler characterization

The rung values sit at the bottom of the spectrum, and the sets attaining them form the *zoo*: at $n = 7$, radius $V = 50$, exactly 29 sets have gap $\frac{2}{15}$, and at $n = 8$, $V = 50$, exactly 21 sets have gap $\frac{2}{17}$. Every zoo member ever observed decomposes as a *core* (a set whose own gap exceeds the rung value) plus one *filler* speed, and the empirical classification of zoos ran through the question: which fillers does a given core admit? Theorem 2.4 answers it in closed form. This section proves the theorem and works the central example.

6.1 Geometry of the core

Fix the core S and the rational target $g \in (0, \frac{1}{2})$, and write $g_S(t) = \min_{s \in S} \|st\|$, a continuous piecewise-linear function. Each sawtooth $\|st\|$ has breakpoints at $t = \text{odd}/(2s)$ and crosses the level $g = p/q$ exactly at the rationals $t = (qm \pm p)/(sq)$. Let $\tau_0 < \tau_1 < \dots < \tau_K$ be the sorted union of $\{0, 1\}$, all breakpoints and all level crossings of all sawtooths. On each open segment (τ_i, τ_{i+1}) every $\|st\|$ is linear and does not cross level g , so the sign of $g_S - g$ is constant there. The escape set $E = \{g_S > g\}$ is therefore a finite union of maximal open intervals (α_j, β_j) ; merging across an event point τ is legitimate only when $g_S(\tau) > g$, so each α_j, β_j carries $g_S = g$ exactly, i.e. $\alpha_j, \beta_j \in L(S)$. Neither endpoint can be 0 or 1, because $g_S(t) \rightarrow 0$ as $t \rightarrow 0$ or 1; the level set $L(S)$ is finite, being contained in the crossing set of the individual sawtooths (no $\|st\|$ is constant on an interval unless two distinct speeds $s \neq s'$ have $\|st\| = \|s't\| = g$ on an interval, which forces $(s - s')t \in \mathbb{Z}$ on an interval and hence $s = s'$).

Proof of Theorem 2.4. Put $h(t) = \min(g_S(t), \|xt\|)$, so that $\text{gap}(S \cup \{x\}) = \max_{[0,1]} h$.

The bound gap $\leq g$ is condition (A). Outside E the core already has $g_S \leq g$. On an escape interval (α, β) the filler must carry the bound: $\|xt\| \leq g$ for all $t \in (\alpha, \beta)$. Writing the circle $\mathbb{R}/\mathbb{Z}$ as the union of the closed windows $[m - g, m + g]$, $m \in \mathbb{Z}$, of length $2g$ separated by open gaps of length $1 - 2g > 0$, the image interval $[x\alpha, x\beta]$ (for $x > 0$) is covered by the windows if and only if it lies inside a single window $[m_j - g, m_j + g]$: a connected interval cannot straddle a gap. By continuity the same holds on the closed $[\alpha, \beta]$, and this is exactly (A).

Equality needs a touch (B). Assume (A), so $\text{gap} \leq g$. The maximum of the continuous h is attained at some t_0 . If $h(t_0) = g$, then $g_S(t_0) \geq g$ and $\|xt_0\| \geq g$. If $g_S(t_0) > g$ then $t_0 \in E$; inside an escape interval, the window condition gives $\|xt_0\| = \text{dist}(xt_0, m_j)$, and an interior point at window distance exactly g would force an endpoint past the window edge, contradicting (A) — formally, $xt_0 = m_j \pm g$ with $x\alpha \leq xt_0 \leq x\beta$ and $x\alpha, x\beta \in [m_j - g, m_j + g]$ forces $xt_0 \in \{x\alpha, x\beta\}$, an endpoint — and $\|xt\| < g$ throughout the open interval would give $h < g$ there. Hence an interior touch cannot occur, the touch must be at an endpoint (where $g_S = g$), or $g_S(t_0) = g$ holds outright. Either way $t_0 \in L(S)$ and $\|xt_0\| \geq g$: this is (B). Conversely, if $t \in L(S)$ has $\|xt\| \geq g$, then $h(t) = \min(g, \|xt\|) = g$, so $\text{gap} \geq g$, and with (A) equality follows.

Degenerate cores. If $\text{gap}(S) < g$ then $L(S) = \emptyset$ and $h \leq g_S \leq \text{gap}(S) < g$: no fillers. If $\text{gap}(S) = g$ then $E = \emptyset$ and (A) is vacuous; $\text{gap}(S \cup \{x\}) = \max \min(g_S, \|x \cdot\|)$ equals g exactly when some t has $g_S(t) = g$ and $\|xt\| \geq g$, i.e. (B) alone; the solutions form infinite arithmetic families since $\|xt\| \geq g$ at a fixed rational t excludes only finitely many residue classes.

Radius. If $E \neq \emptyset$, condition (A) applied to the longest escape interval gives $x(\beta - \alpha) \leq 2g$ for that interval, hence $x \leq 2g / \max_j(\beta_j - \alpha_j)$. $\square$

Example 6.1 (the hub H_2). Take $n = 7$, $g = \frac{2}{15}$, and the core

$$H_2 = (1, 4, 5, 6, 7, 11), \quad \text{gap}(H_2) = \frac{2}{13}.$$

The escape set consists of two mirror intervals,

$$E(H_2) = \left(\frac{32}{105}, \frac{14}{45}\right) \cup \left(\frac{31}{45}, \frac{73}{105}\right), \quad \beta - \alpha = \frac{2}{315},$$

and the level set is

$$L(H_2) = \left\{ \frac{32}{105}, \frac{14}{45}, \frac{7}{15}, \frac{8}{15}, \frac{31}{45}, \frac{73}{105} \right\},$$

of which $\frac{7}{15}$ and $\frac{8}{15}$ are *isolated* level times (local maxima of g_{H_2} at height exactly $\frac{2}{15}$, where the speeds 4 and 11 bind). The radius is $R = \lfloor \frac{4/15}{2/315} \rfloor = 42$. Evaluating (A) on the escape intervals leaves the candidates $\{3, 10, 13, 16, 26, 39\}$ from $\{1, \dots, 42\}$; the touch (B) at $t = \frac{7}{15}$ requires $\|7x/15\| \geq \frac{2}{15}$, i.e. $7x \not\equiv 0, \pm 1 \pmod{15}$, which prunes exactly $x = 13$ ($91 = 6 \cdot 15 + 1$). The theorem yields

$$X(H_2) = \{3, 10, 16, 26, 39\},$$

matching brute force on $[1, 70]$ exactly, with no filler beyond the provable radius $R = 42$: the five fillers generate the five zoo classes $(1, 3, 4, 5, 6, 7, 11)$, $(1, 4, 5, 6, 7, 10, 11)$, $(1, 4, 5, 6, 7, 11, 16)$, $(1, 4, 5, 6, 7, 11, 26)$, $(1, 4, 5, 6, 7, 11, 39)$ of the $n = 7$ census. Note that $26 \equiv 11 \pmod{15}$ lands in the binder class itself: the extension by 26 binds with three speeds $\{4, 11, 26\}$ at $t^* = \frac{7}{15}$.

Corollary 6.2 (fillers bind at level times). *If $\text{gap}(S \cup \{x\}) = g$ and t^* is an argmax, then $g_S(t^*) = g$, i.e. $t^* \in L(S)$; in particular a filler never binds at a time where the core is strictly above g (the window argument forbids $\|xt\| = g$ inside an escape interval), and whenever the filler binds, the core is at level g there as well.*

This is the precise sense in which the theorem is “Lemma A in reverse”: the generalized binding-pair lemma states that an argmax at value $g = \frac{2}{2n+1}$ lives on a grid $d = (2n+1)e$ with binders v, w satisfying $va \equiv \pm 2e \pmod{d}$ (so $v + w \equiv 0 \pmod{d}$); the level times of a core at height g are exactly where such residues can occur, and condition (B) is the residue-cleanliness of the filler there.

6.2 Verification

Theorem 2.4 was verified by exact comparison against brute force: for every core of every zoo member at $n = 5, 6, 7, 8$ (all 612 distinct cores of size $|V| - 1$) and for 480 random cores above the rung, the theorem’s $X(S)$ agrees with the brute-force filler list on $[1, 70]$ with zero mismatches, using two independent implementations of the gap computation (the sweep solver’s integer routine and a Python rational reference) plus a third cross-check. The degenerate regimes (cores on and below the rung) behave as stated; below-rung cores never occur in the zoo regimes, an immediate consequence of the LRC holding for up to seven speeds, since a core of size m has $\text{gap} \geq \frac{1}{m+1} > \frac{2}{2m+3}$.

7 The $n = 8$ census at $V = 50$

The inter-rung question is the target of the hunt: *does any 8-element speed set have gap strictly between $\frac{1}{9}$ and $\frac{2}{17}$?* At $n = 6$ and $n = 7$ the corresponding windows are populated (2 and 4 sets respectively, the $\frac{3}{23}$ counterexamples), so $n = 8$ is the first case where the window could close, and it was verified empty through $V = 38$ ($\binom{38}{8} = 48,903,492$ sets) in the project’s earlier sweep. This section settles the same radius as the $n = 7$ census.

7.1 Method: the probe-lemma filter

The census enumerates all $\binom{50}{8} = 536,878,650$ sets. The feasible cost rests on a one-line certificate: for any fixed probe time t_0 ,

$$\text{gap}(V) \geq \min_{v \in V} \|vt_0\|,$$

because the supremum over t dominates the value at t_0 . The hunt solver (v4) scans the solver's own exact candidate times — peaks odd/ $(2v)$ and crossings $t(v_i \pm v_j) \in \mathbb{Z}$, the complete break-point set of g_V — and skips the vector as soon as one candidate certifies a value above the threshold $\theta = \frac{2}{15}$, which exceeds every hunted band (tight $\frac{1}{9}$, inter-rung $(\frac{1}{9}, \frac{2}{17})$, rung values $\frac{k}{8k+1}$ for $k \leq 6$, all $\leq \theta$). Vectors with $\text{gap} \leq \theta$ — precisely the hunted population — always fall through to the full exact solve. Sampled vectors (every 99,733rd) are always fully solved, preserving sample validation. The filter skipped 99.9985% of vectors; the entire census ran in 215 seconds of single-core time.

Validation of the filtered census: (i) on the full $V = 38$ range, the tight, rung-2 and inter-rung files and the exact bottom spectrum (all 1,565 vectors with $\text{gap} \leq \frac{2}{15}$) are identical to the unfiltered v3 solver's, and the shard-level outputs are byte-identical; (ii) on a $V = 50$ sub-range, filtered and unfiltered runs agree byte-for-byte on all census files and samples (a $105\times$ speedup on that range); (iii) all 5,384 sample records were re-verified against an independent rational implementation; (iv) the accounting is exact: spectrum + filtered = 536,878,650.

7.2 Results

- R1. Tight zoo closed.** $\text{gap} = \frac{1}{9}$ occurs exactly for $\{1, \dots, 8\} \times \{1, 2, 3, 4, 5, 6\}$ — the consecutive family and its scalars, nothing else, at any radius up to 50.
- R2. Rung-2 zoo closed in primitive structure.** $\text{gap} = \frac{2}{17}$ occurs for 21 vectors in exactly the same 7 primitive classes as at $V = 38$, with scalar multiples now up to 4 (Table 1). The seven vectors with $v_8 > 38$ are precisely $\times 2, \times 3, \times 4$ scalings of the existing classes: no new binding structure appears.
- R3. Inter-rung window empty.** No 8-set with speeds up to 50 has gap in $(\frac{1}{9}, \frac{2}{17})$. The spectrum jumps directly from $\frac{1}{9}$ (6 vectors) to $\frac{2}{17}$ (21 vectors); the least attainable value strictly above $\frac{1}{9}$ is the rung-2 value itself.
- R4. Every rung value carries a zoo.** The values $\frac{3}{25}, \frac{4}{33}, \frac{5}{41}, \frac{6}{49}$ are attained by 11, 10, 5 and 3 vectors respectively; the family members $\{1, \dots, 7, 8k\}$ are present in each, and every member satisfies the generalized binding-pair lemma on grids $d = (8k+1)e$ (Table 2). The rung-5 and rung-6 zoos *enter* at this radius, since they need speeds 40 and 48.
- R5. The ladder is certified by the filler theorem.** Each of the seven new rung-2 members decomposes as core + filler with the filler in $X(\text{core})$ computed by Theorem 2.4 at $g = \frac{2}{17}$; all eight 7-subsets of each member support the decomposition. For instance $(3, 6, 9, 12, 15, 18, 21, 48) = 3 \times (1, 2, 3, 4, 5, 6, 7, 16)$ has core $(3, 6, 9, 12, 15, 18, 21)$ with filler $48 \in X(\text{core})$, $R = 84$.
- R6. Feasibility.** Every one of the 536,878,650 vectors is feasible ($\text{gap} \geq \frac{1}{9}$): the Lonely Runner Conjecture holds for 8 speeds up to $V = 50$, extending the $V = 38$ verification.

The bottom of the exact spectrum (all values $\leq \frac{2}{15}$, complete at $V = 50$) reads

value	$\frac{1}{9}$	$\frac{2}{17}$	$\frac{3}{25}$	$\frac{4}{33}$	$\frac{5}{41}$	$\frac{6}{49}$	$\frac{1}{8}$	value	$\frac{7}{55}$	$\frac{6}{47}$	$\frac{5}{39}$	$\frac{4}{31}$	$\frac{3}{23}$	$\frac{8}{61}$	$\frac{5}{38}$
count	6	21	11	10	5	3	763	count	1	17	4	87	275	4	17
								value	$\frac{7}{53}$ and $\frac{2}{15}$						
								count	21 and 1561						

Table 1: The rung-2 zoo at $n = 8$, $V = 50$: 21 vectors, 7 primitive classes (binding pairs modulo 17; grids $d = 17e$ with $e \leq 3$).

primitive class	scalars	binding pair
$A = (1, 2, 3, 4, 5, 6, 7, 16)$	$\times \{1, 2, 3\}$	$\{1, 16\}$
$B = (1, 2, 3, 4, 5, 7, 8, 12)$	$\times \{1, 2, 3, 4\}$	$\{5, 12\}$
$C = (1, 2, 3, 4, 5, 7, 12, 16)$	$\times \{1, 2, 3\}$	$\{1, 16\}$
$D = (1, 2, 5, 6, 7, 8, 11, 13)$	$\times \{1, 2, 3\}$	$\{6, 11\}$
$E = (1, 4, 5, 6, 7, 8, 11, 13)$	$\times \{1, 2, 3\}$	$\{6, 11\}$
$F = (1, 4, 5, 6, 7, 11, 13, 16)$	$\times \{1, 2, 3\}$	$\{1, 16\}, \{4, 13\}, \{6, 11\}$
$G = (1, 5, 6, 7, 8, 11, 13, 21)$	$\times \{1, 2\}$	$\{6, 11\}, \{13, 21\}$

Table 2: Rung- k zoos at $n = 8$, $V = 50$. Each zoo contains the family member $\{1, \dots, 7, 8k\}$ and satisfies the binding-pair lemma on grids $d = (8k + 1)e$.

k	value	vectors	primitive classes	family member present
1	1/9	6	1	yes (consecutive $\times 1..6$)
2	2/17	21	7	yes
3	3/25	11	6	yes
4	4/33	10	10	yes
5	5/41	5	5	yes (new at this radius)
6	6/49	3	3	yes (new at this radius)
inter-rung (1/9, 2/17) 0 (empty at $V \leq 50$)				

The six rung values open the spectrum in exact ladder order, with the family members first; the non-rung values that follow ($\frac{1}{8}, \frac{7}{55}, \dots$) are the next spectral strata.

8 Machine verification

Every theorem in this note is accompanied by an exact machine check; all arithmetic is integer or rational, never floating point.

M1. Rung family, $n = 2..16$, $k = 1..12$ (180 parameter pairs): the gap value equals $\frac{k}{kn+1}$; the argmax set equals $\{\frac{k}{kn+1}, \frac{kn+1-k}{kn+1}\}$ exactly for $k \geq 2$ (including all gcd-degenerate pairs such as $n = k = 3$) and the units grid for $k = 1$; binders are $\{1, kn\}$; every other runner sits at distance $\geq \frac{k+1}{kn+1}$. Zero failures.

M2. Filler characterization: agreement with brute force on all 612 zoo cores ($n = 5..8$), 480 random cores, and the degenerate regimes, as summarized in Section 6.

M3. Census validation: the four validation layers of Section 7 (identity with the unfiltered solver at two radii, byte-level shard identity, 5,384/5,384 independent sample re-verification, exact accounting of all 536,878,650 vectors).

M4. Earlier checks carried over: the $k = 2$ case of Theorem 2.2 (value, unique argmax, binders, slack) was previously machine-verified for $n = 2..40$ with two independent proofs; the present $k \geq 3$ extension and the removal of the gcd condition are new here.

9 Open problems

O1. The $n = 8$ inter-rung theorem. The window $(\frac{1}{9}, \frac{2}{17})$ is empty through $V = 50$ (5.4×10^8 sets), whereas $n = 6$ and $n = 7$ have inter-rung populations at 3/23 and 3/23-type values

living on grids unrelated to $2n+1$. Is the window empty at $n = 8$ for *all* radii? A structural proof — for instance via the binding-pair lemma with modulus 17 — would establish the “second-rung conjecture” at $n = 8$ and would be the first such closure beyond the cases where it is known to fail.

- O2. Rung-zoo structure.** At every radius tested, the rung- k zoo contains the family member $\{1, \dots, n-1, kn\}$ and decomposes into primitive classes generated from lower-rung cores by fillers. The filler theorem reduces this to a finite check per core; a general structural theorem describing the zoos of the rung family remains open.
- O3. $n = 9$ predictions.** The mechanism predicts: tight zoo $\{1, \dots, 9\} \times c$; rung values $\frac{k}{9k+1} = \frac{1}{10}, \frac{2}{19}, \frac{3}{28}, \dots$; rung-2 zoo from zoo-7/zoo-8 cores plus fillers binding modulo 19 (a prime, so every pair $\{v, 19m - v\}$ is admissible at grid level $e = 1$); inter-rung window $(\frac{1}{10}, \frac{2}{19})$ populated or empty — the first test of O1’s mechanism one rung higher.
- O4. Beyond the family.** The no-wrap induction exploits the structure $\{1, \dots, n-1\} \cup \{kn\}$ through the points $\{0, t, \dots, nt\}$. Which other speed families admit a complete argmax classification, and is the argmax of *every* rational-valued gap attained on a rigid grid (the generalized Lemma A)?

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